Black Hole Singularity Resolution in Recognition Science: Discrete Ledger Extremal States

Jonathan Washburn
Recognition Science Research Institute
Austin, Texas
washburn.jonathan@gmail.com

March 2026

Abstract

We show that black hole singularities are dissolved in Recognition Science (RS). The discrete ledger has a minimum voxel size ℓ_0 and minimum tick τ_0, preventing infinite compression. The classical singularity ($r = 0$, $\rho = \infty$) is replaced by an *extremal ledger state*: a region where the recognition density saturates at $\rho_{\max} = 1/\ell_0^3$ (one recognition event per voxel). The Planck density $\rho_{\mathrm{Pl}} \approx 5.2 \times 10^{96}$ kg/m^3 is the RS-predicted saturation scale. Inside the horizon, the J-cost curvature is bounded by $J_{\max} = J(\varphi^{n_{\max}})$, where $n_{\max}$ is the maximum rung on the φ-ladder accessible at the core. The Penrose singularity theorem is not violated—its mathematical conclusion (geodesic incompleteness) is reinterpreted: geodesics terminate not at a singularity but at the extremal surface where the discrete structure becomes manifest.

1 Introduction

The Penrose singularity theorem [**?**] proves that, under general assumptions (trapped surface, energy conditions, global hyperbolicity), gravitational collapse produces geodesic incompleteness—a "singularity" where curvature and density diverge. This implies a breakdown of general relativity at the core of black holes.

Proposed resolutions include loop quantum gravity (bounce cosmology [**?**]), string theory (fuzzball conjecture [**?**]), and noncommutative geometry. Each introduces new structures at the Planck scale.

Recognition Science (RS) provides a simpler resolution: the discrete ledger has a built-in minimum length ℓ_0 that prevents infinite compression. No new physics is introduced—the discreteness that resolves the singularity is the same discreteness that produces quantum mechanics (T2 in the RS forcing chain).

RS derives all physics from the Recognition Composition Law (RCL) [**?**]:

$$J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y), \tag{1}$$

which uniquely determines the cost functional $J(x) = \frac{1}{2}(x + x^{-1}) - 1$. The forcing chain (T0–T8) produces discreteness (T2: continuous configurations cannot stabilize under J), the ledger (T3: $J(x) = J(x^{-1})$ forces double-entry), and $D = 3$ spatial dimensions (T8).

2 The Discrete Ledger and Its Limits

Definition 2.1 (Fundamental Scales)**.** The recognition ledger has:

- Minimum length: $\ell_0 > 0$ (one voxel edge).
- Minimum time: $\tau_0 > 0$ (one tick).
- Minimum volume: $V_{\min} = \ell_0^3$.

Definition 2.2 (Maximum Recognition Density)**.** The maximum density of recognition events is one event per voxel:

$$\rho_{\max} = \frac{1}{\ell_0^3}. \tag{2}$$

Theorem 2.3 (Density Bound)**.** *For any physical configuration in the ledger:* $\rho \leq \rho_{\max}$.

Proof. Each voxel can host at most one recognition event per tick (the Pauli-like exclusion on the ledger). A voxel of volume ℓ_0^3 contributes at most $1/\ell_0^3$ to the density. Subdividing below ℓ_0 is impossible: the discrete lattice has no substructure. □

Remark 2.4 (Status of the exclusion principle). The "Pauli-like exclusion" (one recognition event per voxel per tick) is a structural axiom of the ledger ontology, motivated by the double-entry constraint $J(x) = J(x^{-1})$ (T3) which forces each ledger cell to carry a unique, balanced recognition entry. Multiple simultaneous entries in a single voxel would require the same physical degree of freedom to carry more than one value simultaneously—violating the uniqueness of the recognition state. A rigorous derivation of this exclusion from the RCL alone is an identified open problem; the bound $\rho \leq \rho_{\max}$ is currently a structural postulate consistent with the RS forcing chain.

3 The Classical Singularity

In general relativity, the Schwarzschild solution has

$$\rho(r) \to \infty \quad \text{as } r \to 0, \tag{3}$$

and the Kretschmann scalar $K = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \sim r^{-6}$ diverges. The geodesic incompleteness proved by Penrose means that infalling worldlines terminate in finite proper time.

Remark 3.1. The singularity theorem proves geodesic incompleteness under GR's assumptions. It does not prove that *physical* density diverges—that conclusion requires the additional assumption that spacetime is a smooth continuum at all scales. RS rejects this assumption.

4 The Extremal Ledger State

Definition 4.1 (Extremal State)**.** The extremal ledger state is the configuration in which every voxel in the core region hosts a recognition event at the maximum φ-rung $n_{\max}$:

$$x_i = \varphi^{n_{\max}} \quad \text{for all voxels } i \text{ in the core.} \tag{4}$$

The maximum rung $n_{\max}$ is set by the total mass-energy of the black hole: $M_{\mathrm{BH}} = N_{\mathrm{core}} \cdot m_e \cdot \varphi^{n_{\max}}$, where N_{core} is the number of voxels in the core region. For a solar-mass black hole, $n_{\max} \approx 137$ (the fine-structure-constant rung); for a stellar-mass BH, $n_{\max}$ is of this order.

Theorem 4.2 (Finite Core Cost)**.** *The J-cost at the core is bounded:*

$$J_{\max} = J(\varphi^{n_{\max}}) = \frac{1}{2}\left(\varphi^{n_{\max}} + \varphi^{-n_{\max}}\right) - 1 < \infty. \tag{5}$$

Proof. $\varphi^{n_{\max}}$ is a finite positive real number (the φ-ladder has a maximum rung determined by the total mass-energy of the black hole). J evaluated at any finite positive argument is finite. □

Theorem 4.3 (No Singularity)**.** *In Recognition Science, no physical quantity diverges at the center of a black hole:*

(i) Density: $\rho \leq \rho_{\max} = 1/\ell_0^3 < \infty$.

(ii) Curvature: $K \leq K_{\max} \sim 1/\ell_0^4 < \infty$.

(iii) J-cost: $J \leq J_{\max} < \infty$.

Proof. All three bounds follow from the discreteness of the ledger. (i) Theorem **??**. (ii) Curvature is a second derivative of the metric, which on the lattice has minimum step ℓ_0; hence the maximum second difference is $\sim 1/\ell_0^2$ per spatial direction, giving $K \sim 1/\ell_0^4$. (iii) Theorem **??**. □

5 Reinterpretation of the Penrose Theorem

Theorem 5.1 (Geodesic Termination)**.** *The Penrose singularity theorem's conclusion (geodesic incompleteness) is reinterpreted in RS: infalling geodesics terminate not at a singularity of infinite curvature but at the* extremal surface *where the continuous-geometry approximation breaks down and the discrete ledger structure becomes manifest.*

Remark 5.2. The extremal surface is located at $r_{\mathrm{ext}} \approx \ell_0 \sim \ell_{\mathrm{Pl}}$ (the precise SI value of ℓ_0 depends on the CODATA calibration anchor $\hbar = E_{\mathrm{coh}} \cdot \tau_0$). This is the RS analog of the "Planck star" [**?**] in loop quantum gravity, but requires no new quantization procedure—the discreteness is built into the ledger.

6 The Planck Density

Theorem 6.1 (Saturation Scale)**.** *The maximum physical density in RS is*

$$\rho_{\max} = \frac{1}{\ell_0^3} \sim \rho_{\mathrm{Pl}}, \tag{6}$$

where $\rho_{\mathrm{Pl}} = c^5/(\hbar G^2) \approx 5.2 \times 10^{96}$ kg/m^3 *is the Planck density. The precise numerical coefficient depends on the calibration* $\ell_0 \sim \ell_{\mathrm{Pl}}$*, but the saturation density is of order* 10^{95}–10^{97} *kg/m*^{3} *regardless.*

Remark 6.2. The saturation density being of order the Planck density is consistent with the expectation that new physics (in this case, the discrete ledger) becomes dominant near the Planck scale. This is analogous to the Hagedorn temperature in string theory, where the density of states prevents arbitrarily high temperatures.

7 Comparison with Other Approaches

Approach	**Resolution Mechanism**	**Free Params**
GR (classical)	Singularity (unresolved)	0
Loop QG	Quantum bounce	Barbero–Immirzi γ
String theory	Fuzzball / stretched horizon	String scale
Noncommutative geom.	Smeared source	θ
RS (ledger)	**Extremal ledger state**	**0**

8 Predictions

1. **No trans-Planckian physics**: No experiment will detect length scales below $\ell_0 \sim \ell_P$.

2. **Extremal remnant**: Black hole evaporation terminates at a Planck-mass remnant in the extremal state, not at zero mass. The remnant has mass $M_{\mathrm{rem}} \sim M_{\mathrm{Pl}} \approx 2.2 \times 10^{-8}$ kg and entropy $S_{\mathrm{rem}} \sim O(1)$.

3. **Gravitational wave echoes**: The extremal surface at r_{ext} may produce post-merger gravitational wave echoes. The echo delay time is $\Delta t \sim R_s \ln(R_s/\ell_0)/c$, which for stellar-mass black holes gives $\Delta t \sim O(1)$ ms (in principle detectable by future gravitational wave observatories [**?**]).

4. **Finite core entropy**: The core has finite entropy $S_{\mathrm{core}} \leq k_R \cdot (r_{\mathrm{ext}}/\ell_0)^2$, bounded by the number of extremal voxels.

9 Conclusion

Black hole singularities are dissolved by the discrete ledger. The classical singularity ($r = 0$, $\rho = \infty$) is replaced by an extremal ledger state at $r \approx \ell_0$, where density, curvature, and

J-cost are all bounded and finite. The Penrose theorem is not violated but reinterpreted: geodesics terminate at the extremal surface where the continuum approximation ceases to be valid. The resolution requires zero free parameters—it is a structural consequence of the same discreteness that produces quantum mechanics.

References

[1] J. Washburn, "Recognition Science: Foundations," RS Axioms Document (2026).

[2] R. Penrose, "Gravitational Collapse and Space-Time Singularities," Phys. Rev. Lett. **14**, 57 (1965).

[3] A. Ashtekar, T. Pawlowski, and P. Singh, "Quantum Nature of the Big Bang," Phys. Rev. Lett. **96**, 141301 (2006).

[4] S. D. Mathur, "The Fuzzball Proposal for Black Holes: An Elementary Review," Fortschr. Phys. **53**, 793 (2005).

[5] C. Rovelli and F. Vidotto, "Planck Stars," Int. J. Mod. Phys. D **23**, 1442026 (2014).

[6] J. Abedi, H. Dykaar, and N. Afshordi, "Echoes from the Abyss: Tentative Evidence for Planck-Scale Structure at Black Hole Horizons," Phys. Rev. D **96**, 082004 (2017).

[7] S. W. Hawking, "Breakdown of Predictability in Gravitational Collapse," Phys. Rev. D **14**, 2460 (1976).